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5 March 1991
SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

CHUCKS, INDEPENDENT TYPE, FOUR-JAW

The General Services Administration has authorized the use of this commercial item description.

Abstract. This description covers wrench operated, independent type, four-jaw chucks of the standard, medium duty, commercial type. The chucks are used in a production environment on engine lathes or similar machines to facilitate holding workpieces of irregular shapes where extreme accuracy is required. Chucks covered by this description shall mount directly on or by means of an adapter plate to the machine spindle.

Classification. The chucks shall be of the following types, classes, styles, sizes, and spindle nose types and sizes. The type, class, style, size, and spindle nose type and size required shall be as specified (see ordering data).

Type I - Solid reversible jaws.
Type II - Two-piece reversible jaws.

Class 1 - Forged or cast steel material.
Class 2 - Ductile iron material.

Style A - Flange nose chuck has two rows of tapped holes, or one row of tapped holes and a driving button.

Style B - Flange nose chuck is exactly the same as style A, except the outer holes are clearance holes only, or the inner holes are omitted.

Style D - Camlock spindle nose chuck is equipped with equally spaced clearance holes and each hole is provided with a cam.

Style L - Long taper spindle nose chuck is provided with a key for driving, and a long, steep taper for locating and centering chucks.

Sizes 6 through 42 (see table I).

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis TN 38114-5051.

AMSC N/A FSC 3460

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TABLE I. Chuck size and spindle nose type and size.

Chuck size, diameter (inches)	ANSI Standard spindle nose type and size									
	Style A		Style B		Style D	Style L				
6	A1	A2	B1	B2	D1	L00	L0			
8	A1	A2	B1	B2	D1	L00	L0	L1		
10	A1	A2	B1	B2	D1	L00	L0	L1		
12	A1	A2	B1	B2	D1	-	L0	L1	L2	
14	A1	A2	B1	B2	D1	-	L0	L1	L2	
15	A1	A2	B1	B2	D1	-	L0	L1	L2	
16	A1	A2	B1	B2	D1	-	L0	L1	L2	
18	A1	A2	B1	B2	D1	-	L0	L1	L2	L3
20	A1	A2	B1	B2	D1	-	-	L1	L2	L3
21	A1	A2	B1	B2	D1	-	-	L1	L2	L3
22	A1	A2	B1	B2	-	-	-	L1	L2	L3
24	A1	A2	B1	B2	D1	-	-	-	L2	L3
25	A1	-	B1	B2	D1	-	-	-	L2	L3
28	A1	A2	B1	B2	D1	-	-	-	L2	L3
30	A1	A2	B1	B2	D1	-	-	-	-	L3
32	A1	A2	B1	B2	D1	-	-	-	-	L3
36	A1	A2	B1	B2	D1	-	-	-	-	L3
42	A1	A2	B1	B2	D1	-	-	-	-	L3

NOTE: Chucks shall conform to class I, medium duty, in accordance with ANSI B5.8.

Salient characteristics.

1. Design. The chuck shall be new and of the manufacturer's standard commercial design in effect on the date of the solicitation. The chuck shall be designed to mount directly on, or with the use of an adapter mounting plate onto lathes or other machines having spindle nose styles A, B, D, or L and associated sizes in accordance with ANSI B5.9. Each jaw shall be adjusted separately with its own adjusting screw to facilitate holding workpieces. The chuck shall meet the requirements specified herein for medium duty in ANSI B5.8 and conform to current industrial practices.

2. Materials. All parts and materials shall be of a quality necessary to produce a chuck to meet all requirements specified herein.

3. Chuck body. The body shall be machined to accommodate the jaws, adjusting screws, thrust bearings, and mounting bolts. The chuck body dimensions, T-slots, bolt openings, and eyebolt hole and hardness shall be in accordance with ANSI B5.8. The face of the chuck shall have concentric ring grooves to aid in centering round workpieces.

4. Threads. All threaded parts shall conform to FED-STD-H28.

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5. Chuck jaws. The type I chuck shall have four jaws which shall be of one piece construction, individually adjustable, and reversible for external and internal gripping. All jaws shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 56. Each jaw shall have two step bites and an end bite. The step bites and the end bite shall be serrated for all chuck sizes over 8 inches in diameter. Type II chucks shall have four two-piece (master jaw and top jaw) tongue and groove type jaws in accordance with ANSI B5.8. The master and solid reversible jaws shall be made of alloy steel and the top jaws of steel shall be hardened to not less than Rockwell "C" 56. The top jaw shall have an end bite and two step bites and shall be of the reversible type. The top jaw shall be attached to the master jaw with two socket head cap screws. The step bites and end bite of the top jaw shall be serrated. The bottom of the master jaws shall be provided with precision translating threads to accurately mate with the threads of the jaw adjusting screws.

6. Adjusting screws. The chuck shall have four adjusting screws precision machined to mate with the threads of the corresponding chuck jaws. All adjusting screws shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 52. The outer end of each adjusting screw shall be shaped to accept the applicable chuck wrench. All adjusting screws shall operate smoothly without binding or excessive play. Safety springs shall be required to eject the chuck wrench from the jaw screws.

7. Chuck wrench. Unless otherwise specified (see ordering data), a chuck wrench of the proper size and design shall be furnished with each chuck. The wrench shall be of the T-handle design for manual adjustment of the chuck jaws. The chuck wrench shall be made of steel and shall be hardened to not less than Rockwell "C" 40.

8. Thrust bearings. The chuck jaw shall have thrust bearings which shall be made of alloy steel and shall be hardened to not less than Rockwell "C" 45.

9. Eye bolt. One or more eye bolts for lifting the chuck shall be furnished with each chuck size 12 inches in diameter and larger. The eye bolt specifications shall be in accordance with ANSI B5.8.

10. Mounting screws. The required number of socket head screws or bolts for mounting the chuck to the machine spindle or attaching the chuck to an adapter plate shall be furnished with the applicable chuck.

11. Runout tolerance. With the chuck mounted on a spindle having a spindle nose runout of not more than 0.0003 inch total indicator reading (TIR), the chuck runout shall be within 0.003 inch on the face and periphery during one complete rotation of the spindle. With a test bar having a diameter equal to the through hole in the chucks, mounted within the jaws, the runout shall be not more than 0.003 inch TIR in 360 degree rotation at 3 inches from the edge of the jaws.

12. Workmanship. The workmanship of the chucks shall be of a quality equal to that of comparable products furnished to the commercial market.

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13. Chuck size (diameter). Unless otherwise specified (see ordering data), the diameter of the chuck to be furnished shall be selected from table I.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. During the manufacture of the chucks, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract order.

Quality assurance.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

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Responsibility for compliance. All items shall meet the requirements specified herein. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. Sampling inspection shall be performed in accordance with MIL-STD-105. Sampling shall be at level II with an AQL of 4.0.

Examination. Each chuck submitted for sample inspection shall be visually examined to determine conformance with all requirements of this specification.

Dimensional inspection. Each of the sample chucks shall be inspected to assure that all dimensions are within the tolerances specified herein. Each sample chuck shall be inspected to determine conformance to the runout tolerance specified in paragraph 11 of salient characteristics.

Hardness test. After the sample inspection has been completed, a hardness test shall be performed in accordance with ASTM E-18 on each of the test samples to determine conformance with the hardness requirements of paragraphs 3, 5, 6, 7 and 8 of the salient characteristics. If any of the test samples fail to meet the applicable hardness requirements specified herein, the total lot from which the samples were selected shall be rejected.

NOTES.

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.

b. Type, class, style, chuck size (diameter), spindle nose type and size and quantity required.

c. Specify number, proper size and design of chuck wrenches required.

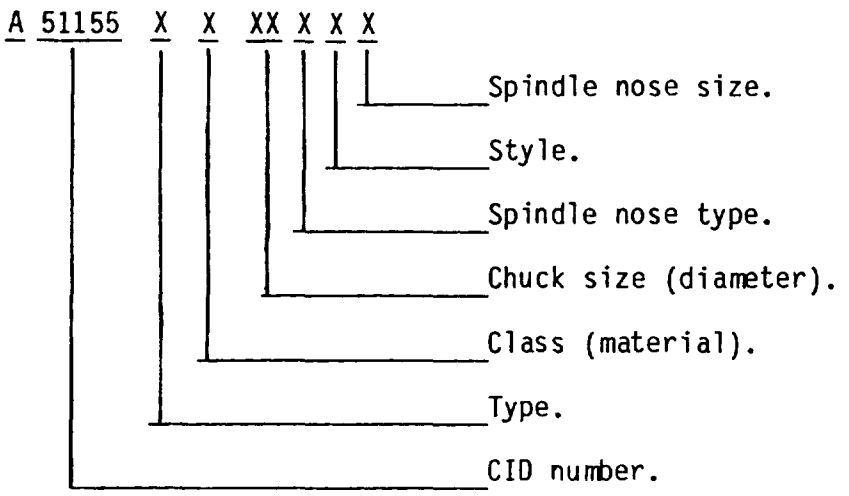
d. Spindle nose size on which the chuck is to be mounted.

e. Spindle nose type on which the chuck is to be mounted.

f. Other requirements.

CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals (see ordering data).

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Federal documents PPP-T-1150, FED-STD-H28 and FED-STD-376 are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.

ANSI B5.8 and ANSI B5.9 are available from the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.

ASTM E 18 is available from American Society for Testing and Materials, 1916 Race street, Philadelphia, PA 19103-1187.

MILITARY INTEREST:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activities

Navy - AS
Air Force - 84
DLA - GS

User Activities

Navy - MC, YD

CIVIL AGENCY COORDINATION ACTIVITY:

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-51149A	91 03 05

3. DOCUMENT TITLE

CHUCKS, INDEPENDENT TYPE, FOUR-JAW

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	(1) Commercial (901)775-6694	(2) AUTOVON AV 683-6694
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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